

ISRO 2025: Topic-Wise Practice Questions

TOPIC 4: THEORY OF COMPUTATION

Based on 2023 ISRO Questions: 7, 8, 55, 56, 57, 58, 59, 60

****Coverage Analysis****: This topic had 8 questions (10%) in 2023 paper, focusing heavily on finite automata, regular expressions, context-free grammars, and language properties.

****Key Concepts Tested****: NFA/DFA conversions, regular expressions, context-free grammars, language hierarchy, computability theory.

FINITE AUTOMATA & REGULAR LANGUAGES (Questions 1-20)

****1. Consider the DFA with alphabet {a,b}, states {q0,q1,q2}, q0 initial, q2 final. $\delta(q0,a)=q1$, $\delta(q0,b)=q0$, $\delta(q1,a)=q2$, $\delta(q1,b)=q0$, $\delta(q2,a)=q2$, $\delta(q2,b)=q2$. Language accepted:****

- (a) Strings containing 'aa'
- (b) Strings ending with 'aa'
- (c) Strings with even number of a's
- (d) Strings starting with 'aa'

****2. Regular expression $(a|b)^*abb$ represents:****

- (a) Strings ending with 'abb'
- (b) Strings containing 'abb'
- (c) Strings starting with 'abb'
- (d) Strings with 'abb' in middle

****3. Number of states in minimal DFA for language $L = \{w \mid w \text{ has even number of a's and odd number of b's}\}$:**

- (a) 2
- (b) 3
- (c) 4
- (d) 6

****4. NFA can be converted to equivalent DFA using:****

- (a) Subset construction method
- (b) Thompson's construction
- (c) McNaughton-Yamada construction
- (d) Brzowski's method

****5. ϵ -NFA with n states can be converted to DFA with at most:****

- (a) n states
- (b) 2^n states
- (c) n^2 states
- (d) n! states

****6. Regular expression $a^*b^*c^*$ generates:****

- (a) $\{a_i b_i c^k \mid i, j, k \geq 0\}$
- (b) $\{a_i b_i c^k \mid i=j=k\}$
- (c) $\{a_i b_i c_i \mid i \geq 0\}$
- (d) $\{abc\}^*$

****7. Which is NOT regular:****

- (a) $\{a^n b^n \mid n \geq 0\}$
- (b) $\{a^n b^{2n} \mid n \geq 0\}$
- (c) $\{a^n b^m \mid n, m \geq 0\}$
- (d) Both (a) and (b)

****8. Pumping lemma for regular languages states:****

- (a) All sufficiently long strings can be pumped
- (b) Some strings cannot be pumped
- (c) Only finite strings can be pumped
- (d) Pumping preserves regularity

****9. Closure properties of regular languages include:****

- (a) Union, intersection, complement
- (b) Concatenation, Kleene star
- (c) Reverse, homomorphism
- (d) All of above

****10. DFA minimization uses:****

- (a) Myhill-Nerode theorem
- (b) Partition refinement
- (c) Equivalence classes
- (d) All of above

****11. Two states are equivalent if:****

- (a) They have same transitions
- (b) They lead to same final states for all strings
- (c) They are both final or both non-final
- (d) They have same labels

****12. Regular expression $(a+b)^* a (a+b)^3$ represents strings:****

- (a) Ending with 'a' followed by exactly 3 symbols
- (b) Having 'a' at 4th position from end
- (c) Containing 'a' somewhere
- (d) Starting and ending with 'a'

****13. Language $\{ww^R \mid w \in \{a,b\}^*\}$ is:****

- (a) Regular
- (b) Context-free but not regular
- (c) Context-sensitive but not context-free
- (d) Not context-sensitive

****14. Finite automaton with output is:****

- (a) Moore machine
- (b) Mealy machine
- (c) Both (a) and (b)
- (d) Neither (a) nor (b)

****15. Moore machine output depends on:****

- (a) Current state only
- (b) Current input only
- (c) Current state and input
- (d) Previous state

****16. Mealy machine output depends on:****

- (a) Current state only
- (b) Current input only
- (c) Current state and input
- (d) Next state

****17. Regular expression for strings over $\{a,b\}$ with no two consecutive a's:****

- (a) $(b^*ab^*)^*b^*$
- (b) $(b|ab)^*a?$
- (c) $b^*(ab^*)^*a?$
- (d) All of above

****18. Language accepted by NFA with ϵ -transitions for token ID starting with letter:****

- (a) $\text{letter}(\text{letter}|\text{digit})^*$
- (b) $(\text{letter}|\text{digit})^*\text{letter}$
- (c) $\text{digit}(\text{letter}|\text{digit})^*$
- (d) $(\text{letter}|\text{digit})^*$

****19. Regular languages are closed under:****

- (a) Homomorphism
- (b) Inverse homomorphism
- (c) Substitution
- (d) All of above

****20. Decision problems for regular languages:****

- (a) Membership - decidable
- (b) Emptiness - decidable
- (c) Equivalence - decidable
- (d) All of above

**CONTEXT-FREE LANGUAGES & GRAMMARS (Questions 21-35)**

****21. Context-free grammar $G: S \rightarrow aSb \mid T, T \rightarrow bT \mid \epsilon$ generates:****

- (a) $\{a^n b^m \mid n \leq m\}$
- (b) $\{a^n b^m \mid n \geq m\}$
- (c) $\{a^n b^{n+m} \mid n, m \geq 0\}$
- (d) $\{a^n b^m \mid m > n\}$

****22. Grammar $S \rightarrow SS \mid aSb \mid bSa \mid \epsilon$ is:****

- (a) Ambiguous
- (b) Unambiguous
- (c) Left-recursive
- (d) Right-recursive

****23. String "aabb" has how many parse trees for grammar $E \rightarrow E+E \mid E^*E \mid id$:****

- (a) 1
- (b) 2
- (c) 3
- (d) More than 3

****24. Left-recursive grammar can cause:****

- (a) Infinite recursion in parsing
- (b) Ambiguity
- (c) Non-termination
- (d) Both (a) and (c)

****25. Chomsky Normal Form requires productions of type:****

- (a) $A \rightarrow BC$ or $A \rightarrow a$
- (b) $A \rightarrow aB$ or $A \rightarrow a$
- (c) $A \rightarrow Ba$ or $A \rightarrow a$
- (d) $A \rightarrow aBc$

****26. Greibach Normal Form requires productions of type:****

- (a) $A \rightarrow BC$
- (b) $A \rightarrow a\alpha$ where α is string of variables
- (c) $A \rightarrow a$
- (d) $A \rightarrow aBc$

****27. CYK algorithm requires grammar in:****

- (a) Chomsky Normal Form
- (b) Greibach Normal Form
- (c) Any form
- (d) Canonical form

****28. Pushdown automaton uses:****

- (a) Stack
- (b) Queue
- (c) Multiple stacks
- (d) Finite memory

****29. Language $\{a^n b^n c^n \mid n \geq 0\}$ is:****

- (a) Regular
- (b) Context-free
- (c) Context-sensitive but not context-free
- (d) Recursively enumerable only

****30. Context-free languages are closed under:****

- (a) Union
- (b) Concatenation
- (c) Kleene star
- (d) All of above

****31. Context-free languages are NOT closed under:****

- (a) Intersection
- (b) Complement
- (c) Both (a) and (b)
- (d) Union

****32. Pumping lemma for CFLs: For CFL L , $\exists p$ such that $|s| \geq p$ implies $s = uvwxy$ with:****

- (a) $|vx| > 0$, $|vwx| \leq p$
- (b) $|vx| \geq 0$, $|vwx| \leq p$
- (c) $|vx| > 0$, $|vwx| > p$
- (d) $|v| = |x|$

****33. DPDA (Deterministic PDA) accepts:****

- (a) All context-free languages
- (b) Deterministic context-free languages
- (c) Regular languages only
- (d) Context-sensitive languages

****34. LR(k) grammars can be parsed by:****

- (a) DPDA
- (b) NPDA
- (c) DFA
- (d) TM

****35. Ambiguous CFG always has:****

- (a) Multiple parse trees for some string
- (b) Left recursion
- (c) ϵ -productions
- (d) Useless symbols

**TURING MACHINES & COMPUTABILITY (Questions 36-50)**

****36. Turing machine consists of:****

- (a) Infinite tape, finite control, tape head
- (b) Finite tape, infinite control
- (c) Stack and finite control
- (d) Queue and finite control

****37. Church-Turing thesis states:****

- (a) Turing machines are most powerful
- (b) Any effectively computable function is Turing computable

- (c) All problems are decidable
- (d) $P = NP$

****38. Halting problem is:****

- (a) Decidable
- (b) Undecidable
- (c) Semi-decidable
- (d) NP-complete

****39. Language is recursive if:****

- (a) Accepted by some TM
- (b) Accepted by TM that halts on all inputs
- (c) Accepted by NPDA
- (d) Context-free

****40. Language is recursively enumerable if:****

- (a) Accepted by some TM
- (b) Accepted by TM that always halts
- (c) Decidable
- (d) Context-sensitive

****41. If L is recursive, then $\bar{L}$ (complement) is:****

- (a) Recursive
- (b) Recursively enumerable
- (c) Neither
- (d) May or may not be recursive

****42. Which statement is FALSE:****

- (a) Every recursive language is RE
- (b) Every finite language is recursive
- (c) If L and $\bar{L}$ are RE, then L is recursive
- (d) Every RE language is recursive

****43. Rice's theorem applies to:****

- (a) All TM properties
- (b) Non-trivial semantic properties of TM languages
- (c) Syntactic properties only
- (d) Decidable properties only

****44. Post Correspondence Problem is:****

- (a) Decidable
- (b) Undecidable
- (c) P-complete
- (d) NP-complete

****45. Universal Turing machine:****

- (a) Can simulate any TM
- (b) Solves halting problem

- (c) Is most efficient TM
- (d) Has infinite states

****46. Decidability of CFL properties:****

- (a) Emptiness - decidable
- (b) Membership - decidable
- (c) Equivalence - undecidable
- (d) All of above

****47. Context-sensitive languages are accepted by:****

- (a) Linear bounded automaton
- (b) Turing machine
- (c) Pushdown automaton
- (d) Both (a) and (b)

****48. Phrase structure grammar generates:****

- (a) Regular languages
- (b) Context-free languages
- (c) Context-sensitive languages
- (d) Recursively enumerable languages

****49. Type-0 grammars have productions of form:****

- (a) $A \rightarrow \alpha$ where A is variable, α is any string
- (b) $\alpha A \beta \rightarrow \alpha \gamma \beta$ where A is variable
- (c) $A \rightarrow aB$ or $A \rightarrow a$
- (d) No restrictions

****50. Language hierarchy (Chomsky):****

- (a) Regular $\subset$ CFL $\subset$ CSL $\subset$ REL
- (b) Regular $\subset$ CSL $\subset$ CFL $\subset$ REL
- (c) CFL $\subset$ Regular $\subset$ CSL $\subset$ REL
- (d) All are equal

**PARSING & COMPILER THEORY (Questions 51-55)**

****51. LL(1) parsing:****

- (a) Left-to-right scan, leftmost derivation, 1 lookahead
- (b) Left-to-right scan, rightmost derivation, 1 lookahead
- (c) Right-to-left scan, leftmost derivation
- (d) Uses bottom-up approach

****52. LR(1) parsing:****

- (a) Left-to-right scan, rightmost derivation, 1 lookahead
- (b) Left-to-right scan, leftmost derivation, 1 lookahead
- (c) Uses recursive descent
- (d) Top-down approach

****53. FIRST(α) contains:****

- (a) Terminals that can begin strings derived from α
- (b) Non-terminals that can begin derivations
- (c) All symbols in α
- (d) Leftmost symbols only

****54. FOLLOW(A) contains:****

- (a) Terminals that can appear immediately after A
- (b) Terminals in productions of A
- (c) All terminals
- (d) Non-terminals after A

****55. Shift-reduce conflict occurs when:****

- (a) Parser can either shift or reduce
- (b) Multiple reduce actions possible
- (c) Multiple shift actions possible
- (d) Grammar is ambiguous

**ANSWER KEY - THEORY OF COMPUTATION**

****1-10:**** (a), (a), (c), (a), (b), (a), (d), (a), (d), (d)

****11-20:**** (b), (b), (b), (c), (a), (c), (d), (a), (d), (d)

****21-30:**** (a), (a), (b), (d), (a), (b), (a), (a), (c), (d)

****31-40:**** (c), (a), (b), (a), (a), (a), (b), (b), (b), (a)

****41-50:**** (a), (d), (b), (b), (a), (d), (d), (d), (a), (a)

****51-55:**** (a), (a), (a), (a), (a)

**Critical Study Focus for September 14th:**

1. ****Finite Automata (Q1-20):**** Master DFA/NFA construction and regular expression conversion
2. ****Context-Free Grammars (Q21-35):**** Focus on grammar types, parse trees, and ambiguity
3. ****Computability (Q36-50):**** Understand decidability concepts and Turing machines
4. ****Parsing (Q51-55):**** Know LL(1) vs LR(1) differences

****Key Concepts to Master:****

- Regular expression to DFA conversion
- Context-free grammar parsing and ambiguity detection
- Decidability vs undecidability problems
- Chomsky hierarchy and language properties

****Study Approach:**** This topic requires conceptual understanding more than calculations. Focus on recognizing patterns and applying theorems correctly.